EXPERIMENT-13

AIM: Study about Wireshark

Wireshark is an open-source packet analyzer, which is used for **education, analysis, software development, communication protocol development, and network troubleshooting**.

It is used to track the packets so that each one is filtered to meet our specific needs. It is commonly called as a **sniffer, network protocol analyzer, and network analyzer**. It is also used by network security engineers to examine security problems.

Wireshark is a free to use application which is used to apprehend the data back and forth. It is often called as a free packet sniffer computer application. It puts the network card into an unselective mode, i.e., to accept all the packets which it receives.

A packet is a unit of data which is transmitted over a network between the origin and the destination. Network packets are small, i.e., maximum **1.5 Kilobytes for Ethernet packets and 64 Kilobytes for IP packets**. The data packets in the Wireshark can be viewed online and can be analyzed offline

Installation of Wireshark Software

Below are the steps to install the Wireshark software on the computer:

- Open the web browser.
- Search for '**Download Wireshark**.'
- Select the Windows installer according to your system configuration, either 32-bt or 64-bit. Save the program and close the browser.
- Now, open the software, and follow the install instruction by accepting the license.
- The Wireshark is ready for use.

On the network and Internet settings option, we can check the interface connected to our computer.

If you are Linux users, then you will find Wireshark in its package repositories.

i. Packet Capture Using Wire shark

Capturing your traffic with Wireshark
After starting Wireshark, do the following:

1. Select **Capture** | **Interfaces**
2. Select the interface on which packets need to be captured. This will usually be the interface where the Packet/s column is constantly changing, which would indicate the presence of live traffic). If you have multiple network interface cards (i.e. LAN card and Wi-Fi adapter) you may need to check with your IT administrator to determine the right interface.

3. Click the **Start** button to start the capture.
4. Recreate the problem. The capture dialog should show the number of packets increasing. Try to avoid running any other internet applications while capturing, closing other browsers, Instant messengers etc.
5. Once the problem which is to be analyzed has been reproduced, click on **Stop**. It may take a few seconds for Wireshark to display the packets captured.
6. Save the packet trace in the default format. Click on the **File** menu option and select **Save As**. By default Wireshark will save the packet trace in libpcap format. This is a filename with a.pcap extension.

ii. Starting Wire shark using command line

You can start Wireshark from the command line, but it can also be started from most Window managers as well. In this section we will look at starting it from the command line.

Wireshark supports a large number of command line parameters. To see what they are, simply enter the command *wireshark –h*

```
Wireshark 3.7.3 (v3.7.3rc0-25-g72703582d587)
Interactively dump and analyze network traffic.
See https://www.wireshark.org for more information.

Usage: wireshark [options] ... [ <infile> ]

Capture interface:
  -i <interface>, --interface <interface>
                           name or idx of interface (def: first non-loopback)
  -f <capture filter>      packet filter in libpcap filter syntax
```

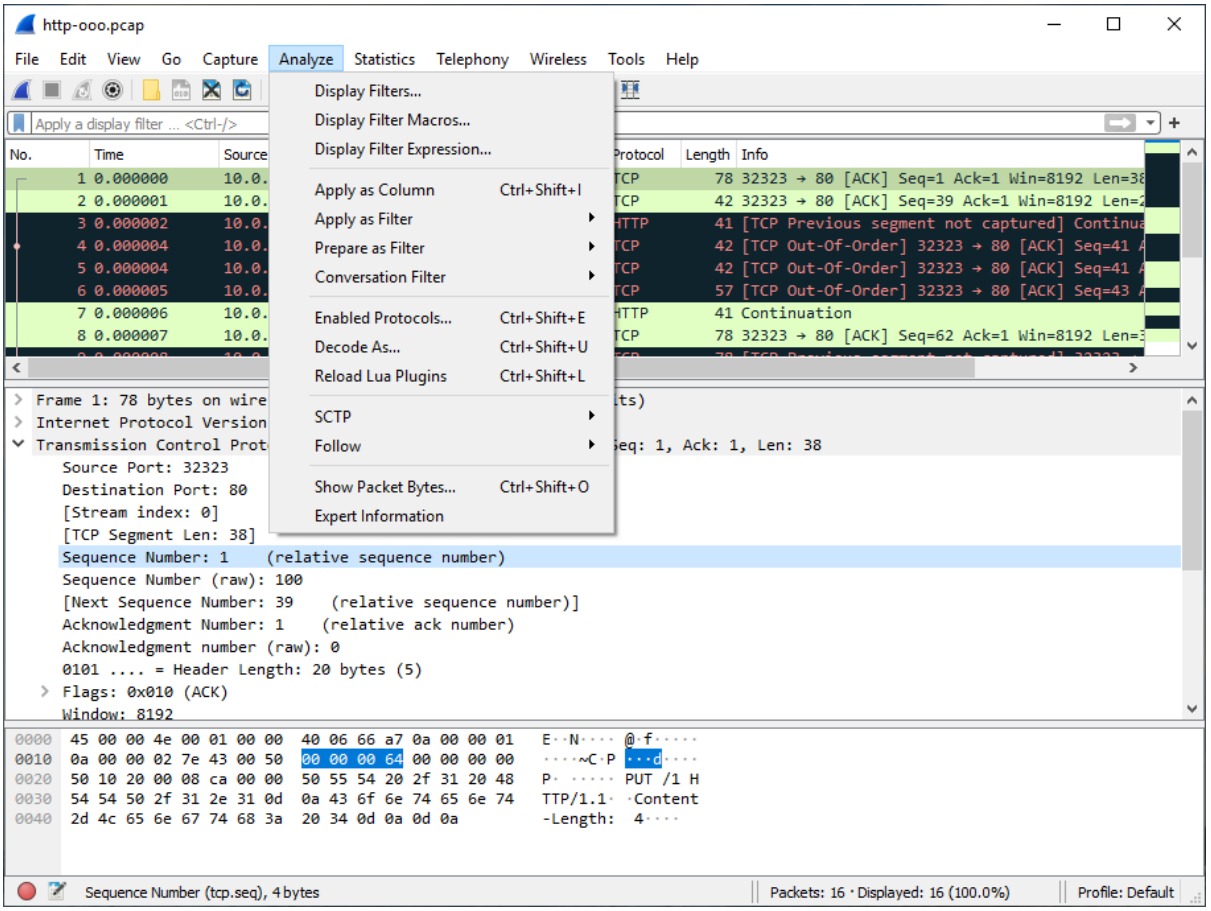
iii. Viewing Captured Traffic

Once you have captured some packets or you have opened a previously saved capture file, you can view the packets that are displayed in the packet list pane by simply **clicking on a packet in the packet list pane**, which will bring up the selected packet in the tree view and byte view panes. Once you have captured some packets or you have opened a previously saved capture file, you can view the packets that are displayed in the packet list pane by simply **clicking on a packet in the packet list pane**, which will bring up the selected packet in the tree view and byte view panes.

iv. Analysis and Statistics & Filters.

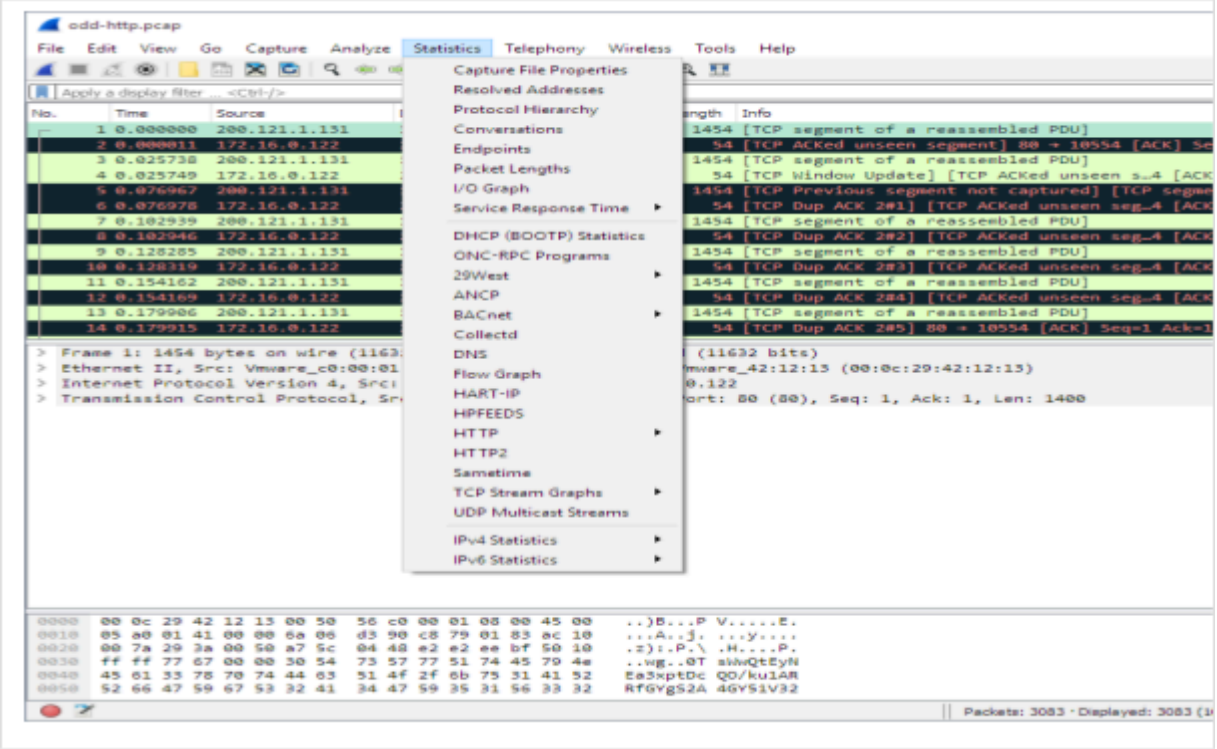
Analysis menu in wireshark

The “Analyze” Menu The Wireshark Analyze menu contains the fields shown in Analyze menu items.



The “Statistics” Menu in wireshark

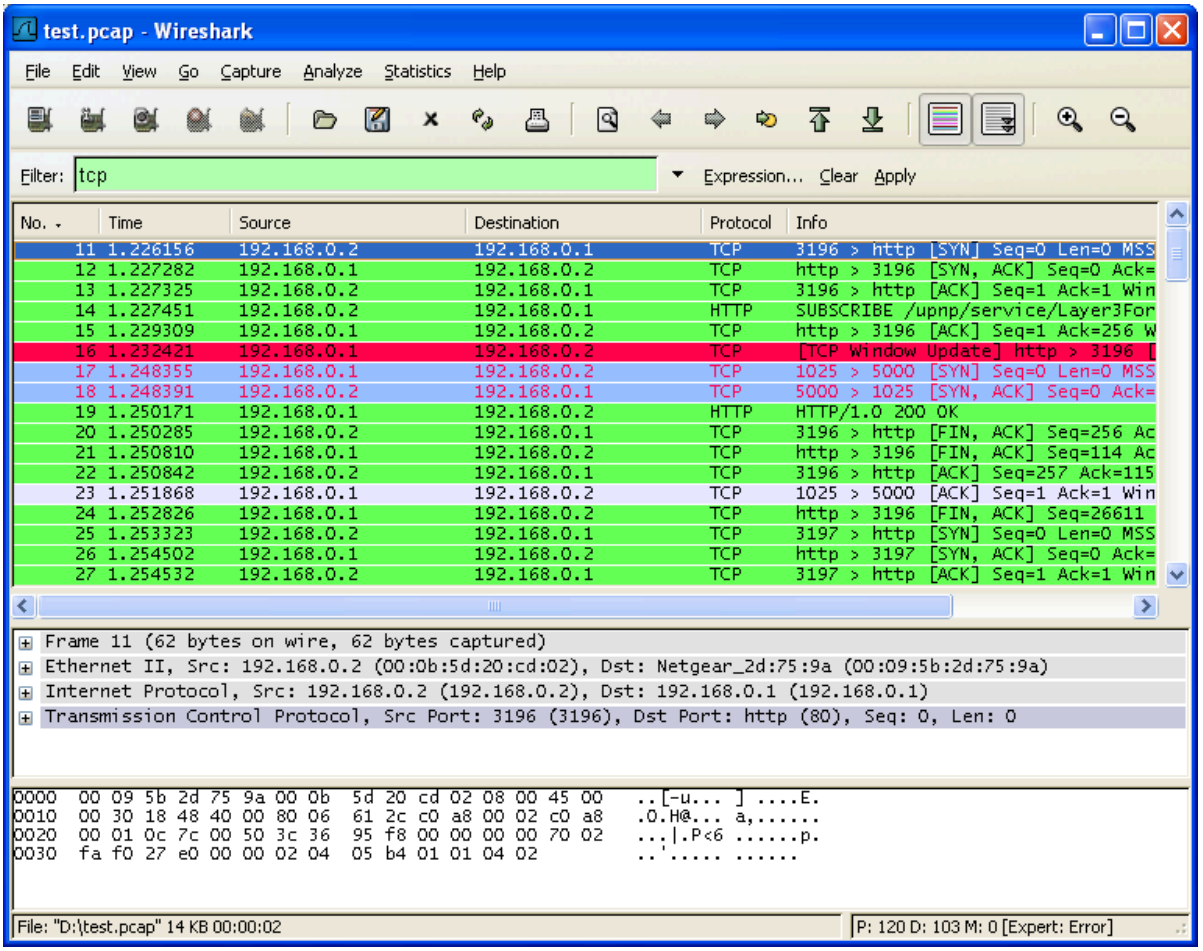
The Wireshark Statistics menu contains the fields



shown in Table 3.9, “Statistics menu items”.

Filters in wireshark

Wireshark has two filtering languages: **capture filters and display filters**. Capture filters are used for filtering when capturing packets, “Filtering while capturing”. Display filters are used for filtering which packets are displayed.


test.pcap - Wireshark
File Edit View Go Capture Analyze Statistics Help
Filter: tcp
Expression... Clear Apply
No. Time Source Destination Protocol Info
11 1.226156 192.168.0.2 192.168.0.1 TCP 3196 > http [SYN] Seq=0 Len=0 MSS
12 1.227282 192.168.0.1 192.168.0.2 TCP http > 3196 [SYN, ACK] Seq=0 Ack=
13 1.227325 192.168.0.2 192.168.0.1 TCP 3196 > http [ACK] Seq=1 Ack=1 Win
14 1.227451 192.168.0.2 192.168.0.1 HTTP SUBSCRIBE /upnp/service/Layer3For
15 1.229309 192.168.0.1 192.168.0.2 TCP http > 3196 [ACK] Seq=1 Ack=256 W
16 1.232421 192.168.0.1 192.168.0.2 TCP [TCP Window Update] http > 3196 [
17 1.248355 192.168.0.1 192.168.0.2 TCP 1025 > 5000 [SYN] Seq=0 Len=0 MSS
18 1.248391 192.168.0.2 192.168.0.1 TCP 5000 > 1025 [SYN, ACK] Seq=0 Ack=
19 1.250171 192.168.0.1 192.168.0.2 HTTP HTTP/1.0 200 OK
20 1.250285 192.168.0.2 192.168.0.1 TCP 3196 > http [FIN, ACK] Seq=256 Ac
21 1.250810 192.168.0.1 192.168.0.2 TCP http > 3196 [FIN, ACK] Seq=114 Ac
22 1.250842 192.168.0.2 192.168.0.1 TCP 3196 > http [ACK] Seq=257 Ack=115
23 1.251868 192.168.0.1 192.168.0.2 TCP 1025 > 5000 [ACK] Seq=1 Ack=1 Win
24 1.252826 192.168.0.1 192.168.0.2 TCP http > 3196 [FIN, ACK] Seq=26611
25 1.253323 192.168.0.2 192.168.0.1 TCP 3197 > http [SYN] Seq=0 Len=0 MSS
26 1.254502 192.168.0.1 192.168.0.2 TCP http > 3197 [SYN, ACK] Seq=0 Ack=
27 1.254532 192.168.0.2 192.168.0.1 TCP 3197 > http [ACK] Seq=1 Ack=1 Win
Frame 11 (62 bytes on wire, 62 bytes captured)
Ethernet II, Src: 192.168.0.2 (00:0b:5d:20:cd:02), Dst: Netgear_2d:75:9a (00:09:5b:2d:75:9a)
Internet Protocol, Src: 192.168.0.2 (192.168.0.2), Dst: 192.168.0.1 (192.168.0.1)
Transmission Control Protocol, Src Port: 3196 (3196), Dst Port: http (80), Seq: 0, Len: 0
0000 00 09 5b 2d 75 9a 00 0b 5d 20 cd 02 08 00 45 00 ..[-u...] ....E.
0010 00 30 18 48 40 00 80 06 61 2c c0 a8 00 02 c0 a8 .0.H@... a,......
0020 00 01 0c 7c 00 50 3c 36 95 f8 00 00 00 00 70 02 ...|.P<6 ......p.
0030 fa f0 27 e0 00 00 02 04 05 b4 01 01 04 02 ..'..... ......
File: "D:\test.pcap" 14 KB 00:00:02
P: 120 D: 103 M: 0 [Expert: Error]